

Touying + Navigator

Structural Transitions via Native Hooks

Background & Motivation

Motivation

Prior work

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Why Presentation Structure Matters

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Structured presentations help audiences track where they are in a talk. Navigator generates transition slides automatically from the document outline.

- Section transitions: show the full plan
- Subsection transitions: highlight current position
- Slides stay in sync without manual effort

Why Presentation Structure Matters

“A well-structured talk is a talk the audience can follow even when they lose concentration for a moment.”

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Existing Approaches and Their Limitations

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Most Typst presentation packages handle navigation manually:

- **Polylux:** `show heading: h => render-transition(h, ...)`
- **Diatypst / typslides:** similar hook-based approach
- **Touying:** native section/subsection hooks → cleaner integration

Design & Implementation

Architecture

Short titles

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Architecture of the Touying Integration

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The key insight: Touying's `new-section-slide-fn` hook fires exactly when a new section or subsection is encountered, and `here()` already points to that heading's location.

This makes `match-page-only: true` reliable for identifying the active entry in `progressive-outline`, without needing `h.location()` explicitly.

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Short titles

Handling Short Titles in the Outline

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Long subsection names would overflow the transition slide. Navigator supports <short> markers placed right after a heading:

== A Very Long and Descriptive Subsection Title

`#metadata("Short version") <short>`

The outline then uses the short version; the slide header keeps the full title.

Handling Short Titles in the Outline

Short titles are resolved per level via `use-short-title: true` in `progressive-outline`. You can also pass a per-level dictionary:

```
use-short-title: ("level-1": false, "level-2": true)
```

Results & Conclusion

Results

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Observed Improvements in Audience Engagement

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In this example, the transition slides show:

- The **active** section/subsection in white
- **Completed** items in semi-transparent white
- **Upcoming** items in dimmed white

The visual hierarchy guides the audience through the structure.

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Conclusion

Conclusion and Future Directions

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**Touying's native hooks + progressive-outline =
zero-boilerplate structural transitions**

See also: `polylux_progressive_outline.typ` for the alternative approach.